

LECTURE 2

Non-Stationarity: Unit Roots and Trends

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Agenda

- 01 Motivation: why stationarity matters
- 02 Random walks and integrated processes
- 03 Spurious regression
- 04 Trend-stationarity vs. difference-stationarity
- 05 Dickey–Fuller and ADF tests
- 06 Phillips–Perron and KPSS tests
- 07 Testing in practice: size, power, pitfalls
- 08 Business applications

We use the log of the S&P 500 index (daily, 2000–2023) and the log of Brazilian CPI (monthly, 1995–2023) to illustrate every concept.

Section 1

Motivation: Why Stationarity Matters

From Stationarity to Non-Stationarity I

Recap: what stationarity gives us

- Time-stable mean and variance
- ACF/PACF identify the model
- OLS is consistent and asymptotically normal
- Standard t - and F -tests are valid
- Forecasts revert to a fixed long-run mean

What breaks down without stationarity

- Variance grows without bound
- No fixed long-run mean to revert to
- OLS t -stats do **not** follow t -distributions
- Spurious regressions between unrelated series
- Standard forecasting intervals explode with horizon

Most macroeconomic and financial series — prices, interest rates, exchange rates, sales volumes — contain stochastic trends. Ignoring non-stationarity invalidates inference, produces phantom relationships, and distorts forecasts.

From Stationarity to Non-Stationarity II

Examples of likely non-stationary series

- Asset prices and stock indices
- Nominal exchange rates
- Price levels and CPI
- GDP and consumption
- Brand equity indices over long horizons
- Interest rate levels (in some regimes)

Examples of likely stationary series

- Asset *returns* (log differences of prices)
- Inflation rates (differenced CPI)
- Interest rate *spreads*
- Detrended productivity growth
- Marketing metrics in short windows

Transform non-stationary series before applying ARMA models. **Test** formally — do not assume.

Section 2

Random Walks and Integrated Processes

The Random Walk I

$$Y_t = Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} (0, \sigma^2), \quad Y_0 = 0.$$

Key properties

- $Y_t = \sum_{s=1}^t \varepsilon_s$ (cumulated shocks)
- $\mathbb{E}(Y_t) = 0, \quad \text{Var}(Y_t) = t\sigma^2$
- Variance **grows linearly** with t
- $\text{Cov}(Y_t, Y_{t+s}) = t\sigma^2$ (depends on t , not just s)
- ACF: $\rho(s) = \sqrt{t/(t+s)} \rightarrow 1$ as $t \rightarrow \infty$

A random walk has $\hat{\rho}(1) \approx 0.99$, but so does a very persistent AR(1) with $\phi = 0.99$. The two **behave very differently** for inference and forecasting. Formal unit root tests are needed.

The Random Walk II

Random walk with drift

$$Y_t = \mu + Y_{t-1} + \varepsilon_t \implies Y_t = \mu t + \sum_{s=1}^t \varepsilon_s$$

- Adds a deterministic **linear trend** on top of stochastic wandering
- $\mathbb{E}(Y_t) = \mu t$, $\text{Var}(Y_t) = t\sigma^2$ — still non-stationary
- Typical in nominal asset prices, log GDP, log CPI

For a stationary process, Wold guarantees an $\text{MA}(\infty)$ representation with square-summable coefficients. For $I(1)$, the coefficients are all unity — shocks have **permanent** effects.

Integrated Processes: $I(d)$

A series Y_t is **integrated of order d** , written $Y_t \sim I(d)$, if it must be differenced d times to achieve stationarity:

$$\Delta^d Y_t = (1 - L)^d Y_t \sim I(0).$$

$I(0)$

Stationary. No differencing needed.

Examples: returns, spreads, residuals of a well-specified model.

$I(1)$

One difference needed.

Examples: log prices, log exchange rates, log CPI, log GDP.

$I(2)$

Two differences needed.

Examples: log nominal wages in some periods; velocity in hyperinflation. Rare in practice.

The AR representation and the unit root

An AR(1) $Y_t = \phi Y_{t-1} + \varepsilon_t$ has characteristic equation $1 - \phi L = 0$.

- $|\phi| < 1$: root **outside** unit circle $\Rightarrow$ stationary I(0)
- $\phi = 1$: root **on** unit circle $\Rightarrow$ I(1) — the **unit root** case
- $|\phi| > 1$: explosive — rarely relevant in practice

AR(p) with a unit root:

$$\phi(L)Y_t = \varepsilon_t \iff (1 - L)\tilde{\phi}(L)Y_t = \varepsilon_t$$

where $\tilde{\phi}(L)$ has all roots **outside** the unit circle.

Testing for a unit root is equivalent to testing whether the AR polynomial has a root *on* the unit circle. If it does, differencing is required before standard inference.

Section 3

Spurious Regression

Spurious Regression: Granger and Newbold (1974) I

If two **independent** random walks Y_t and X_t are regressed on each other, OLS will almost always find a statistically significant relationship — even though none exists.

Simulation setup (Granger & Newbold, 1974):

- Generate $Y_t = Y_{t-1} + \varepsilon_t^Y$ and $X_t = X_{t-1} + \varepsilon_t^X$ independently
- Regress Y_t on X_t : $Y_t = \alpha + \beta X_t + u_t$
- Despite $\text{Corr}(\varepsilon^Y, \varepsilon^X) = 0$, the t -statistic on $\hat{\beta}$ exceeds 1.96 in about **75% of Monte Carlo replications**

Why does this happen?

- Both series share a common time trend (stochastic)
- OLS mistakes this shared non-stationarity for a structural relationship
- R^2 is high but residuals are highly autocorrelated (DW ≈ 0)

In a spurious regression:

- Residuals are non-stationary (ADF on $\hat{u}_t$ fails to reject unit root)
- DW statistic ≈ 0

• $R^2 \rightarrow 1$ as $T \rightarrow \infty$ (Use other checks the relation)

Spurious Regression: Granger and Newbold (1974) II

Business example: spurious marketing regression

Regress monthly *sales volume* on monthly *competitor advertising spend* over 10 years without testing for unit roots:

- Both series may be $I(1)$ (growing over time)
- OLS will find a significant positive relationship
- But the relationship may be entirely due to common growth trends
- Correct approach: test both series, difference if $I(1)$, or find a cointegrating relationship

Whenever $R^2 > DW$ in a time series regression, suspect spurious regression. Always test residuals for stationarity.

Section 4

Trend-Stationarity vs. Difference-Stationarity

Two Types of Trends I

Trend-Stationary (TS)

$$Y_t = \mu + \delta t + u_t, \quad u_t \sim I(0)$$

- Trend is **deterministic**
- Deviations from trend are **temporary**
- Shocks have **no permanent** effect
- Correct treatment: **detrend** (regress on t)
- After detrending: $\hat{u}_t \sim I(0)$

Difference-Stationary (DS)

$$Y_t = \mu + Y_{t-1} + \varepsilon_t$$

- Trend is **stochastic**
- Deviations are **permanent**
- Shocks have a **lasting** effect on the level
- Correct treatment: **difference**
- After differencing: $\Delta Y_t = \mu + \varepsilon_t \sim I(0)$

Two Types of Trends II

Applying the wrong treatment is costly. Detrending a DS process leaves non-stationary residuals (spurious detrending). Differencing a TS process over-differences and introduces non-invertible MA roots.

Two Types of Trends III

Nelson and Plosser (1982) — landmark paper testing 14 US macroeconomic series:

- Applied (then-new) Dickey–Fuller tests to data spanning 1909–1970
- Could **not** reject the unit root for **13 out of 14** series (exception: unemployment rate)
- Conclusion: most macro series are DS, not TS
- Implication: **permanent** effects of monetary and fiscal shocks — contrary to the real business cycle view of purely cyclical fluctuations

If sales, prices, or macro drivers follow DS processes, then demand shocks, policy changes, and structural shifts have **permanent** effects. Models must account for this in long-run forecasting.

Section 5

Dickey-Fuller Tests

The Dickey–Fuller Test I

Subtract Y_{t-1} from both sides of $Y_t = \phi Y_{t-1} + \varepsilon_t$:

$$\Delta Y_t = \pi Y_{t-1} + \varepsilon_t, \quad \pi = \phi - 1.$$

Null: $H_0 : \pi = 0$ (unit root) vs. **Alt:** $H_1 : \pi < 0$ (stationary).

Three standard specifications

- 1 No constant, no trend:

$$\Delta Y_t = \pi Y_{t-1} + \varepsilon_t$$

- 2 Constant, no trend:

$$\Delta Y_t = \alpha + \pi Y_{t-1} + \varepsilon_t$$

- 3 Constant and trend:

$$\Delta Y_t = \alpha + \delta t + \pi Y_{t-1} + \varepsilon_t$$

Under H_0 , the t -statistic on $\hat{\pi}$ does **not** follow a standard t -distribution.

It follows a **Dickey–Fuller distribution** — left-skewed, with critical values more negative than -1.96 :

$$\tau \xrightarrow{d} \frac{\int_0^1 W dW}{\left(\int_0^1 W^2\right)^{1/2}}$$

The Dickey–Fuller Test II

Augmented Dickey–Fuller (ADF) test — Dickey & Fuller (1981); Said & Dickey (1984)

Add lagged differences to control for serial correlation in ε_t :

$$\Delta Y_t = \alpha + \delta t + \pi Y_{t-1} + \sum_{j=1}^p \gamma_j \Delta Y_{t-j} + \varepsilon_t$$

- Test statistic: t -ratio on $\hat{\pi}$ — same DF critical values apply
- Lag length p : select by AIC/BIC, or use sequential testing (start with $p_{\max}$, remove insignificant lags)
- Critical values depend on specification (no constant / constant / trend) and T

Specification	CV (5%)
No constant	−1.95
Constant	−2.86
Constant + trend	−3.41

Too few lags $\Rightarrow$ size distortion (over-reject H_0).
Too many lags $\Rightarrow$ power loss (under-reject H_0).
Use BIC for consistent selection.

Choosing the correct specification

A common practical question: should I include a constant? A trend?

- 1 Look at a **time plot** of the series before testing
- 2 If the series wanders around a fixed level $\Rightarrow$ use *constant, no trend*
- 3 If the series shows a clear upward/downward slope $\Rightarrow$ use *constant and trend*
- 4 If the series oscillates around zero $\Rightarrow$ use *no constant*

Including unnecessary deterministics reduces power but avoids spurious rejections. Under-specifying (omitting a constant when there is a drift) inflates size severely.

Section 6

Phillips–Perron and KPSS Tests

Phillips–Perron (PP) Test I

Motivation: ADF adds lagged differences to control for autocorrelation. PP uses a different approach: non-parametric correction of the t -statistic.

PP test regression

$$\Delta Y_t = \alpha + \pi Y_{t-1} + u_t$$

Run OLS, then correct the $\hat{\pi}$ t -statistic for serial correlation and heteroskedasticity using a **Newey–West long-run variance estimator**:

$$\hat{\omega}^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^q \left(1 - \frac{j}{q+1}\right) \hat{\gamma}_j$$

Advantages over ADF

- No lag length selection needed
- Robust to heteroskedastic errors
- Can handle MA components in errors

PP test has poor finite-sample properties when errors are highly autocorrelated — can have severe size distortions. ADF is generally preferred in practice.

Phillips–Perron (PP) Test II

KPSS Test — Kwiatkowski, Phillips, Schmidt & Shin (1992)

KPSS **reverses** the null and alternative hypotheses:

$$H_0 : Y_t \sim I(0) \quad \text{vs.} \quad H_1 : Y_t \sim I(1)$$

$$Y_t = \xi t + r_t + \varepsilon_t, \quad r_t = r_{t-1} + u_t, \quad u_t \sim \text{iid}$$

Test $H_0 : \sigma_u^2 = 0$ (random walk component has zero variance $\Rightarrow$ stationarity).

Test statistic:

$$\eta = \frac{1}{T^2 \hat{\omega}^2} \sum_{t=1}^T S_t^2$$

where $S_t = \sum_{s=1}^t \hat{e}_s$ are partial sums of OLS residuals.

Use ADF and KPSS **together**:

- Both reject $\Rightarrow$ conflicting evidence
- ADF rejects, KPSS does not $\Rightarrow$ strong evidence for $I(0)$
- ADF does not reject, KPSS rejects $\Rightarrow$

Section 7

Testing in Practice: Pitfalls and Power

Unit root tests have **low power** against stationary alternatives with a root close to unity. A series with $\phi = 0.97$ is stationary, but ADF will rarely reject $H_0 : \phi = 1$ in samples of $T = 100$.

Structural breaks further reduce power — Perron (1989):

- If a stationary series has a structural break (level shift), the ADF test will tend to **not** reject the unit root null
- The break mimics the behaviour of a random walk
- Solution: unit root tests that allow for breaks — Perron (1989), Zivot–Andrews (1992)

Re-testing Nelson–Plosser's 14 series with a break at the Great Crash (1929) and the 1973 oil shock: **11 out of 14** series rejected the unit root — the opposite conclusion from Nelson–Plosser. Breaks matter enormously.

Practical checklist for unit root testing

1. **Plot the series.** Look for trends, breaks, structural shifts, level changes.
2. **Choose specification.** Constant? Trend? Based on the plot, not data-mining.
3. **Select lag length.** Use BIC or sequential t -testing on γ_j in ADF.
4. **Run ADF and KPSS.** Use both for robustness. Check they agree.
5. **Check for structural breaks.** If suspected, use Zivot–Andrews or Perron (1989).
6. **Test at multiple frequencies.** Monthly data may require seasonal unit root tests (HEGY).
7. **Report all specifications.** Do not selectively report only the result that favours your prior.

If in doubt, difference. Over-differencing costs some efficiency but avoids spurious inference. Under-differencing (treating $I(1)$ as $I(0)$) causes much more serious distortions.

Section 8

Business Applications

Finance

Efficient market hypothesis (EMH):

If markets are efficient, log prices follow a random walk (no drift in excess returns). Testing H_0 : unit root in log prices is a test of EMH in this sense.

Interest rate modelling:

Rates can be $I(1)$ or near- $I(1)$. This matters for VaR models, bond pricing, and term structure models.

Pairs trading:

Requires two $I(1)$ series to be cointegrated (see Lecture 4). First step is always unit root testing.

Marketing

Long-run sales modelling:

If sales and advertising spend are both $I(1)$, a static regression will be spurious unless they are cointegrated.

Brand equity:

Brand equity indices constructed over long time horizons are often $I(1)$. Correct differencing is needed before regression analysis.

Demand systems:

Price and quantity indices are typically $I(1)$. Error correction models (Lecture 3) handle this correctly.

R and Python implementation

R:

```
library(tseries); library(urca) adf.test(log(sp500), k=4) ur.df(log(sp500),  
type="trend", lags=4) kpss.test(log(sp500), null="Trend") PP.test(log(sp500))  
za <- ur.za(log(sp500), model="both")
```

Python:

```
from statsmodels.tsa.stattools import adfuller, kpss adf_result = adfuller(  
log_sp500, maxlag=4, regression='ct', autolag='BIC') kpss_result = kpss(  
log_sp500, regression='ct')
```

Core references

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- Phillips, P.C.B. & Perron, P. (1988). Testing for a unit root in time series regression. *Biometrika*, 75, 335–346.
- Kwiatkowski et al. (1992). Testing the null hypothesis of stationarity. *JoE*, 54, 159–178.

Applied papers

- Nelson, C.R. & Plosser, C.I. (1982). Trends and random walks in macroeconomic time series. *JME*, 10, 139–162.
- Perron, P. (1989). The great crash, the oil price shock, and the unit root hypothesis. *Econometrica*, 57, 1361–1401.
- Jacobsen, B. & Zhang, C.Y. (2014). The Halloween indicator. *Finance Research Letters*, 11, 112–120.
- Cerqueira, V. et al. (2020). Evaluating time series forecasting models. *Machine Learning*, 109, 1997–2028.

Software: R (`tseries`, `urca`) or Python (`statsmodels`). Applied exercises: ADF + KPSS on S&P 500, USD/BRL, Brazilian CPI, and a series of your choice from your research area.

Next: Lecture 3

Dynamic Regression, Error Correction, and Exogeneity

ADL models · ECM · Engle–Hendry–Richard exogeneity framework